PROJECT PROPOSAL

Unified EdTech Ecosystem for Classroom Management, Collaboration and Process-Based Academic Integrity Monitoring

1. Project Team

The project is submitted by the following team.

Sr. No.	Name	Roll No.	Role
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2. Short Summary

The project is a multi-platform educational technology (EdTech) ecosystem that brings classroom administration, student collaboration, and academic integrity monitoring together in one unified platform. Rather than scanning finished text for AI-like language, the integrity system analyzes how a piece of work was produced.

2.1 High-Level Technical Architecture

Layer	Primary Technology	Core Functionality
Frontend UI	Flutter (Dart)	Cross-platform mobile and desktop user interface
Native Core	Rust / C++	High-performance telemetry engine, local encryption, and dynamic library
Browser Extension	WebAssembly (Rust/C++) + JavaScript (Manifest V3)	Low-overhead typing capture, paste detection, and web telemetry across all major browsers

Layer	Primary Technology	Core Functionality
Backend & Storage	PostgreSQL & REST/WebSocket API	Relational data persistence, persistent section rosters, and real-time chat sync

3. Detailed Summary

3.1 What Is the Project?

The project is a multi-platform educational technology (EdTech) ecosystem designed to bridge classroom administration, student collaboration, and an academic integrity (AI detection) monitoring system into a single unified platform.

3.2 What Does It Offer?

A. Cross-Platform Classroom Command Center (Flutter UI)

- **Unified Student Portal:** Students can view enrolled courses, inspect assignment briefs, attach files, and submit coursework directly from their mobile or desktop devices.
- **Integrated Academic Calendar & Deadline Alerts:** A synchronized calendar aggregates assignment due dates, quiz schedules, and institutional events. Automated push notifications alert students to upcoming deadlines and notify them immediately if an assignment becomes overdue.
- **Multi-Tiered Communication Hub:** Role-restricted group chat channels separate high-volume student discussions from official announcements.

B. High-Performance Native Telemetry Engine (Rust / C++)

- **Native Shared Engine via FFI & WebAssembly:** On desktop and mobile platforms, the native binary interacts directly with the Flutter frontend through foreign function interfaces (dart:ffi / flutter_rust_bridge). For web browsers, the same core is compiled into WebAssembly (WASM).
- **Real-Time Keystroke & Telemetry Capture:** The native core captures typing cadence, inter-keystroke delays, cursor movement patterns, and paste volume in real time. Because it runs on a low-level compiled engine, it processes tens of thousands of telemetry events per minute with near-zero CPU footprint and no user-perceptible UI latency.
- **Cryptographic Signatures & Privacy:** Telemetry logs are encrypted and cryptographically signed on the local machine using native cryptographic libraries before transmission, ensuring data cannot be tampered with in transit.

C. Process-Based Integrity Browser Extension (Manifest V3)

- **Effort Score Calculation:** Instead of scanning completed text for AI-like language patterns, the extension analyzes how the text was produced. A high effort score is awarded to work created through organic typing and iterative edits, while sudden massive paste events trigger lower effort ratings.
- **AI Chatbot Interaction Tracking:** The extension detects tab switching to external generative AI platforms (e.g., ChatGPT, Claude) during writing sessions and logs copy-paste transfers for transparent review.

- **Time-Lapse Writing Replay:** Instructors can watch a fast-forward visual playback of a student's document construction. A teacher can see an essay take shape in about 30 seconds, verifying research flow, outline development, and organic editing.

D. Teacher AI Bot & Smart Workspace Integration

- **1-Click Google Sheets / Excel Sync:** Attendance logs, grade updates, and submission statuses can be pushed directly to Google Sheets through the Google Sheets API, or exported as CSV/Excel workbooks.
- **Automated Missing Assignment Nudges:** The system monitors real-time submission rosters and lets instructors automatically draft and send personalized reminders to students with pending or overdue work.
- **Classroom Auto-Organizer:** The AI bot assists in structuring assignment rubrics, organizing folder structures for incoming file attachments, and drafting routine class announcements.